

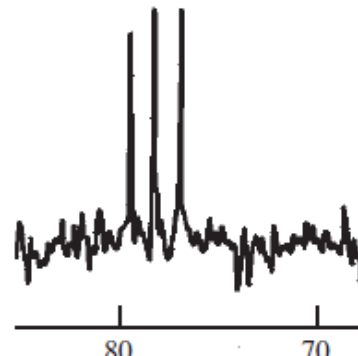
Heteronuclear Coupling of Carbon-13 Fluorine-19 Phosphorous-31

The ^{13}C NMR peaks of
two common solvents. (a) Dimethylsulfoxide- d_6 (b)
 CD_3COCD_3 ,

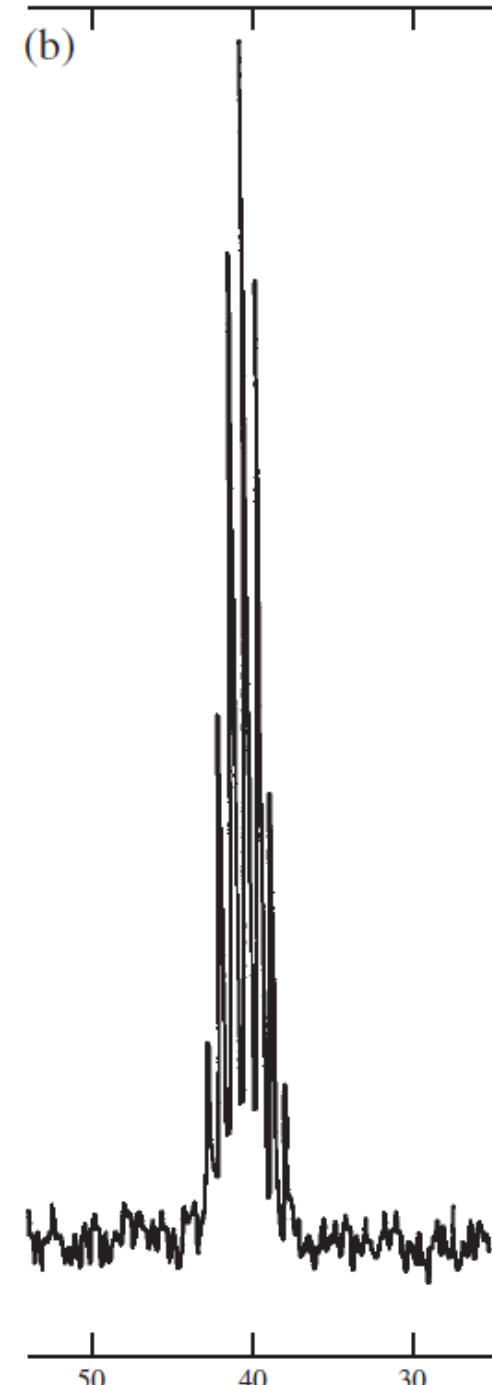
The ^{13}C NMR peaks of
two common solvents. (a) Chloroform-d.
(b) Dimethylsulfoxide-d₆.

Acetone-d₆, $\text{CD}_3\text{I}|\text{CO}|\text{CD}_3$, shows the same ^{13}C septet splitting
pattern as dimethylsulfoxide-d₆

(a)

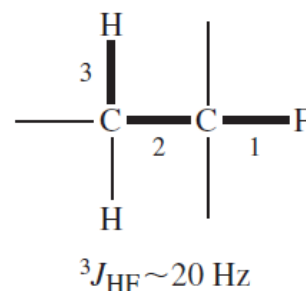
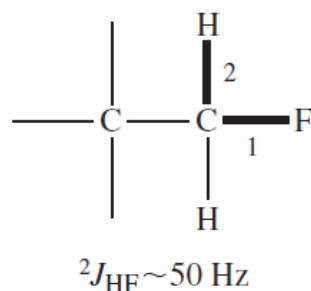


(b)



A. ^1H to ^{19}F Couplings

Coupling constants for ^{19}F to ^1H can be very large, and one may typically expect 2J and 3J couplings values to be much larger than typical ^1H to ^1H coupling constants. The following examples show the effects of 2J and 3J couplings of ^1H to ^{19}F as the proton moves away from the electronegative fluorine atom.



Compound: F-CH₂-CH₂-OH (2-fluoroethanol) in CDCl₃ solvent.

Observed Data:

•¹H NMR:

- 4.50 ppm (2H, dt, J = 47 Hz, 5 Hz)
- 3.70 ppm (2H, dt, J = 25 Hz, 5 Hz)
- 3.4 ppm (1H, broad s)

Assign the peaks and the j values as tree diagram.

Clue: ²J and ³J values of C-F are high and around 50 and 25 hz respectively.

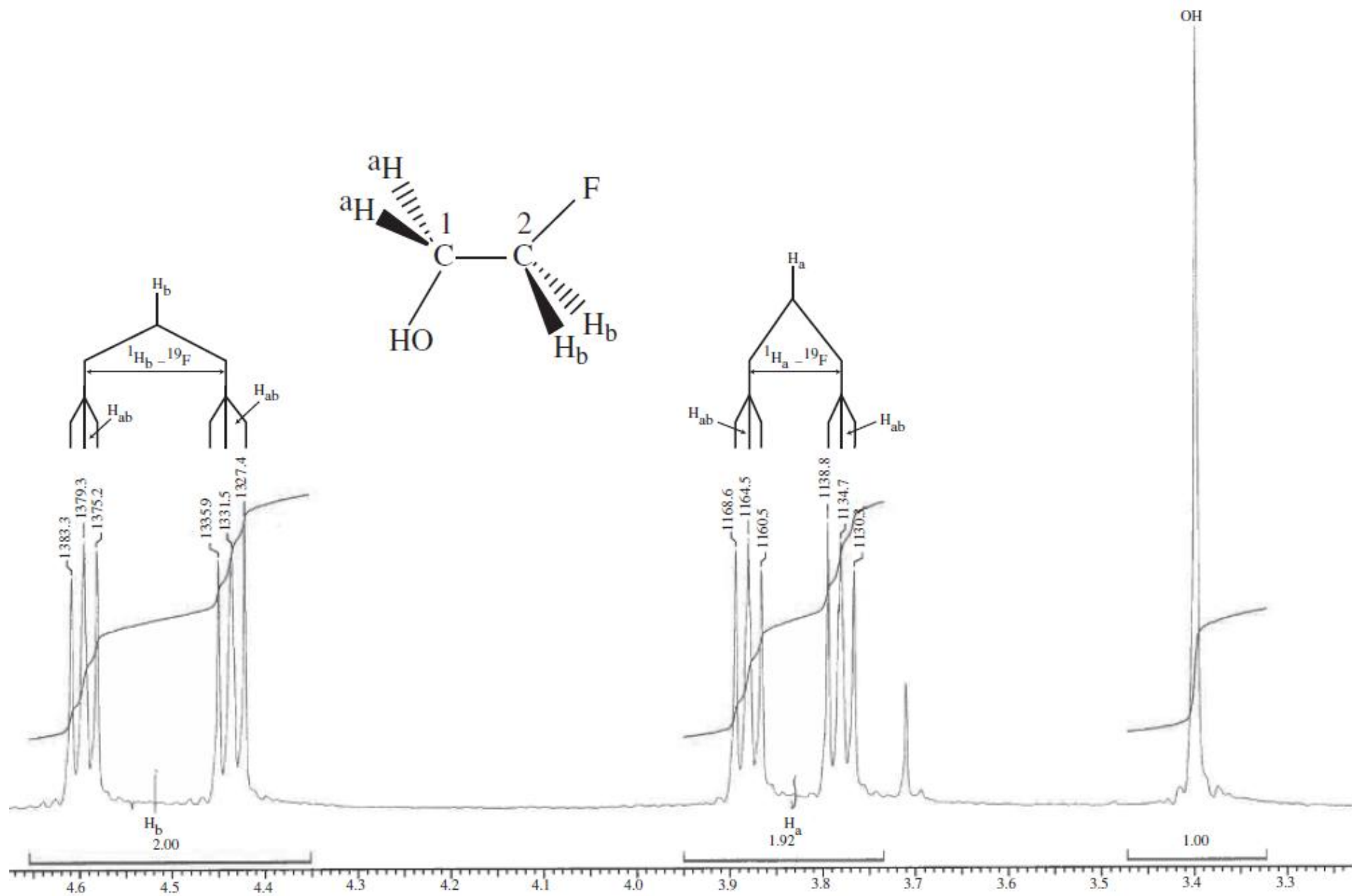
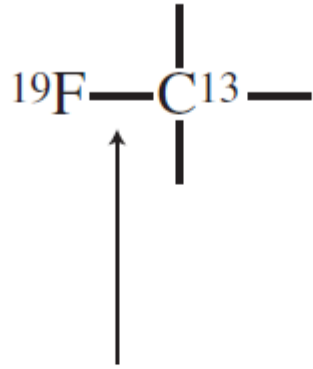
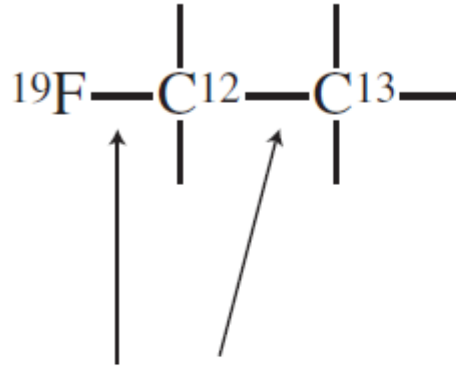


FIGURE 7.73 ^1H NMR Spectrum of 2-fluoroethanol (300 MHz, CDCl_3)



one bond connecting
C to F = 1J

Very high j values
Close to 160-320

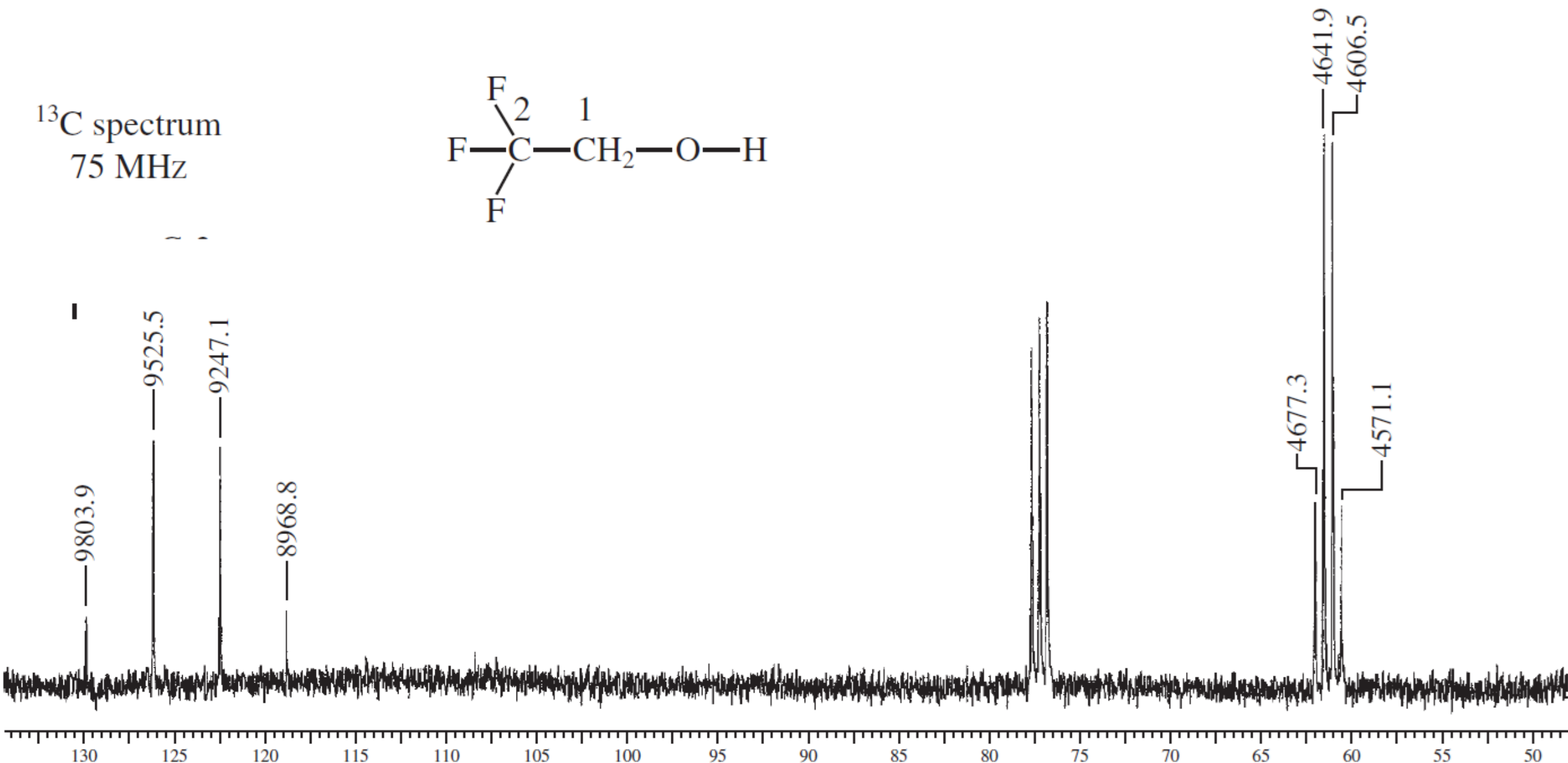
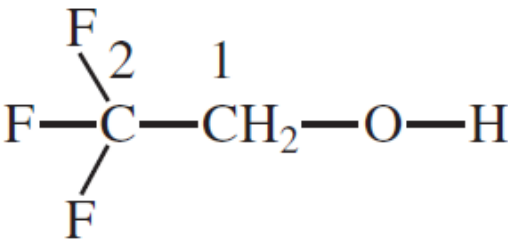


two bonds connecting C to F = 2J
The carbon-13 atom is connected
through the common isotope,
carbon-12, to fluorine-19.

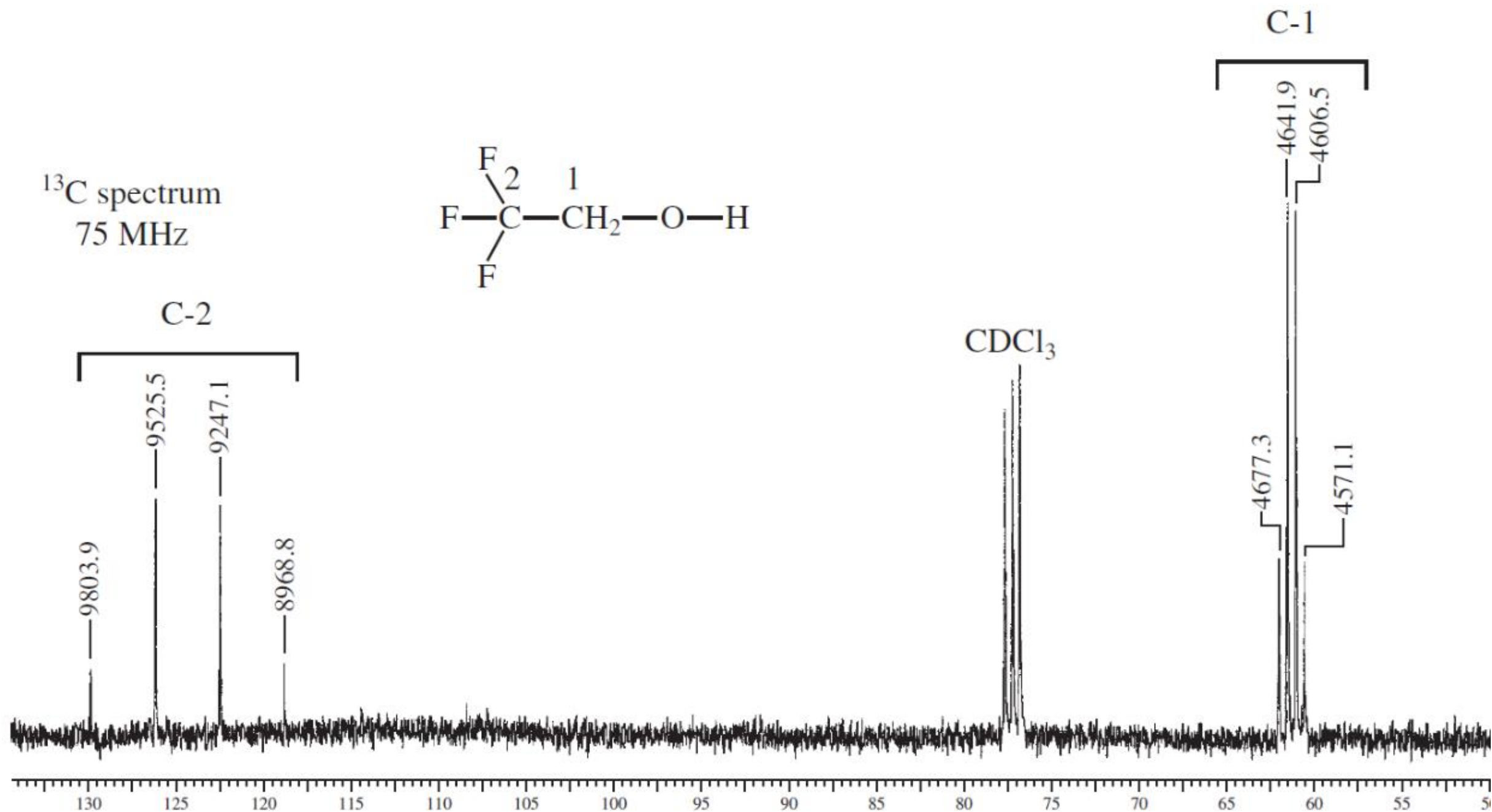
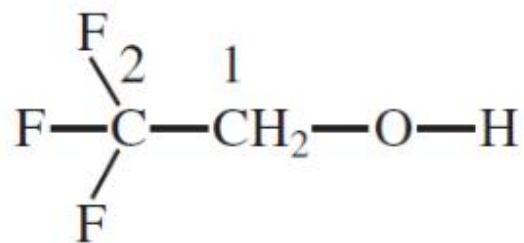
Relatively low, but still j values
Close to up to 40 hz

Can you assign the peaks and explain the splitting in CDCl3? C to F 1J = 200-320Hz and C to F 2J = 40Hz

^{13}C spectrum
75 MHz

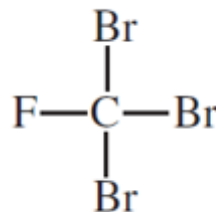


^{13}C spectrum
75 MHz



Heteronuclear Coupling of Carbon-13 to Fluorine-19

How many peaks in ^{13}C do you expect? ^{13}C - ^{19}F (1J): 370 Hz
 ^{19}F is 100% abundance with I value of 1/2



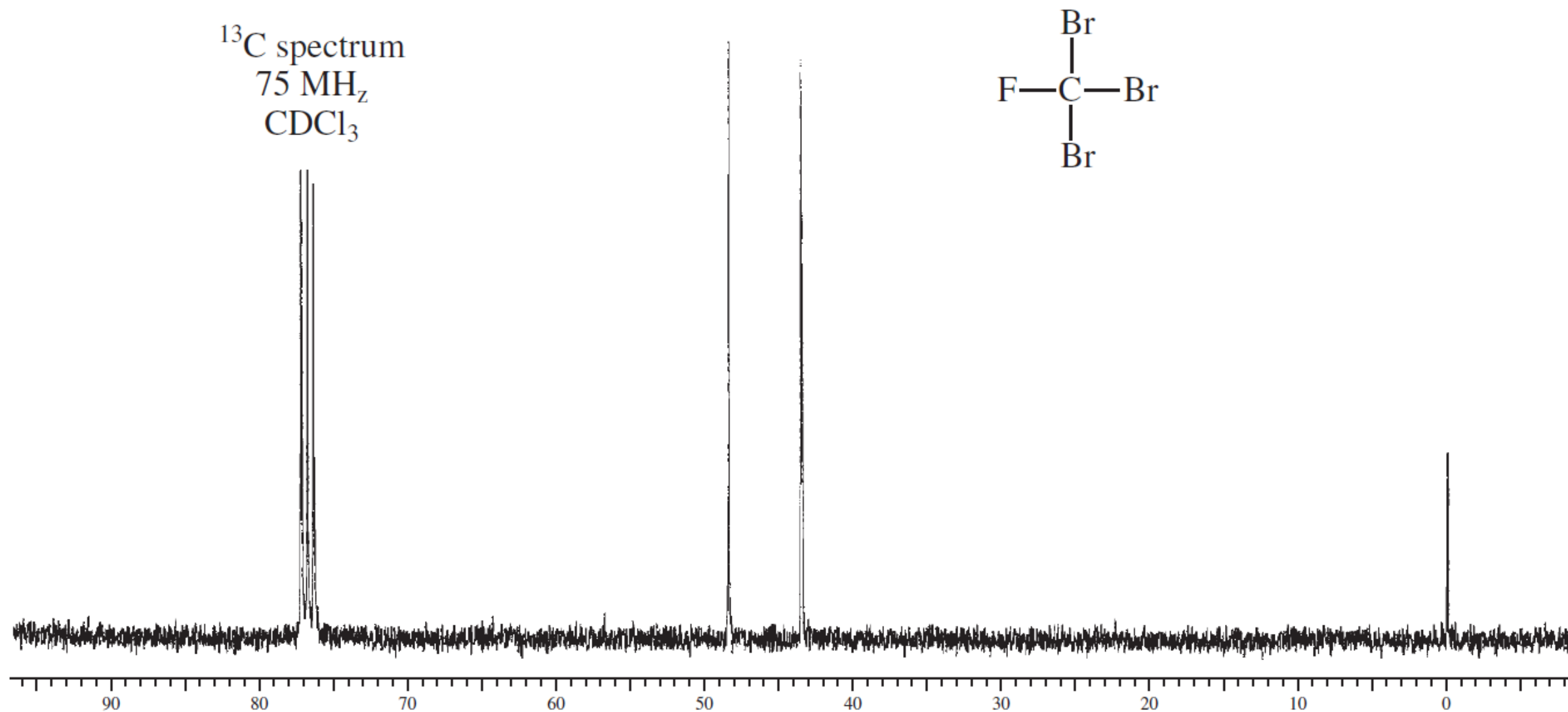
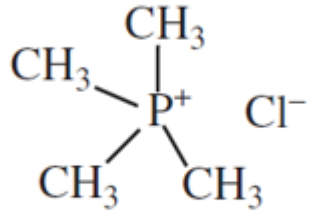


FIGURE 6.21 The ^{13}C proton-decoupled spectrum of CBr_3F (75 MHz).



How many peaks in ^{13}C do you expect? ^{31}P is 100% abundance with I value of 1/2

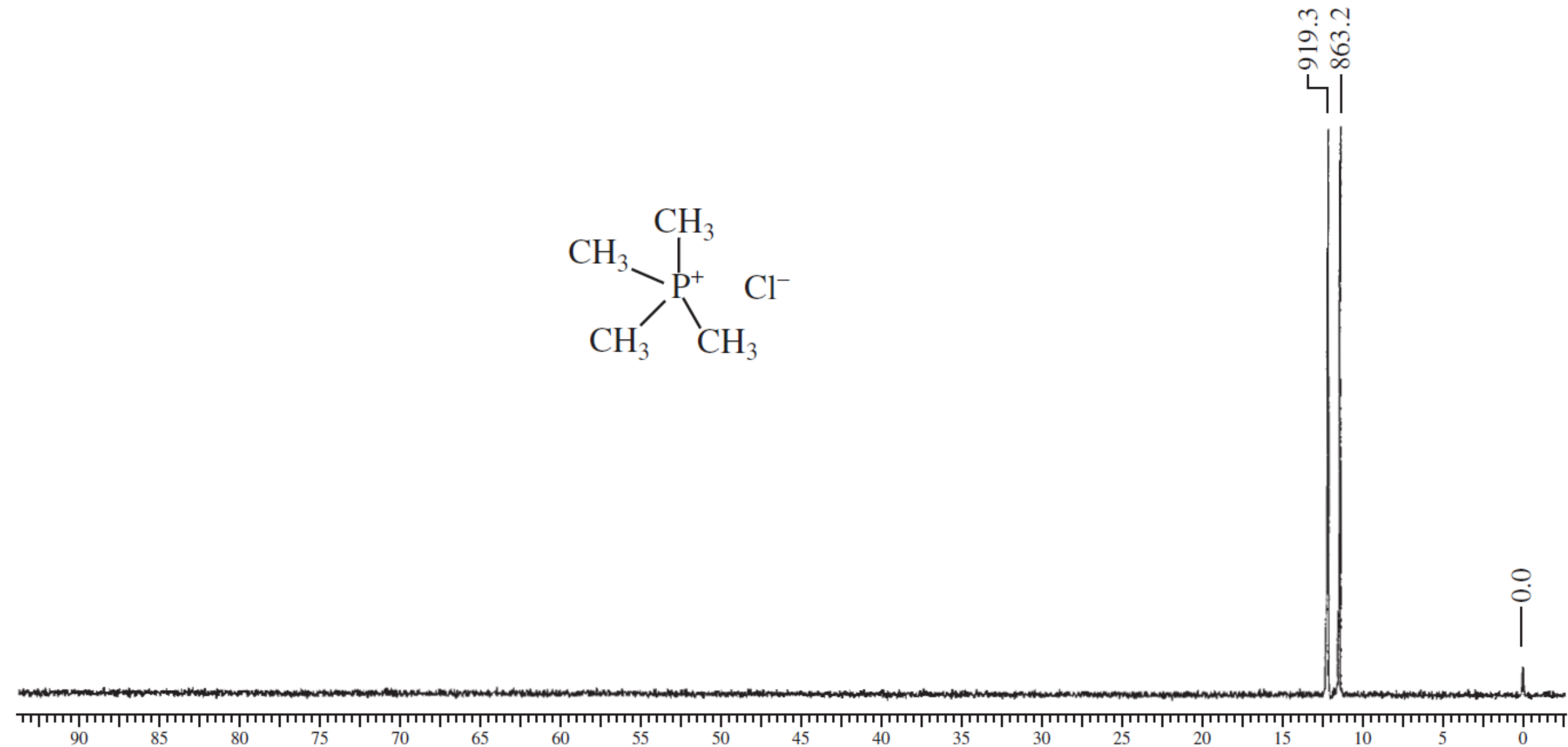
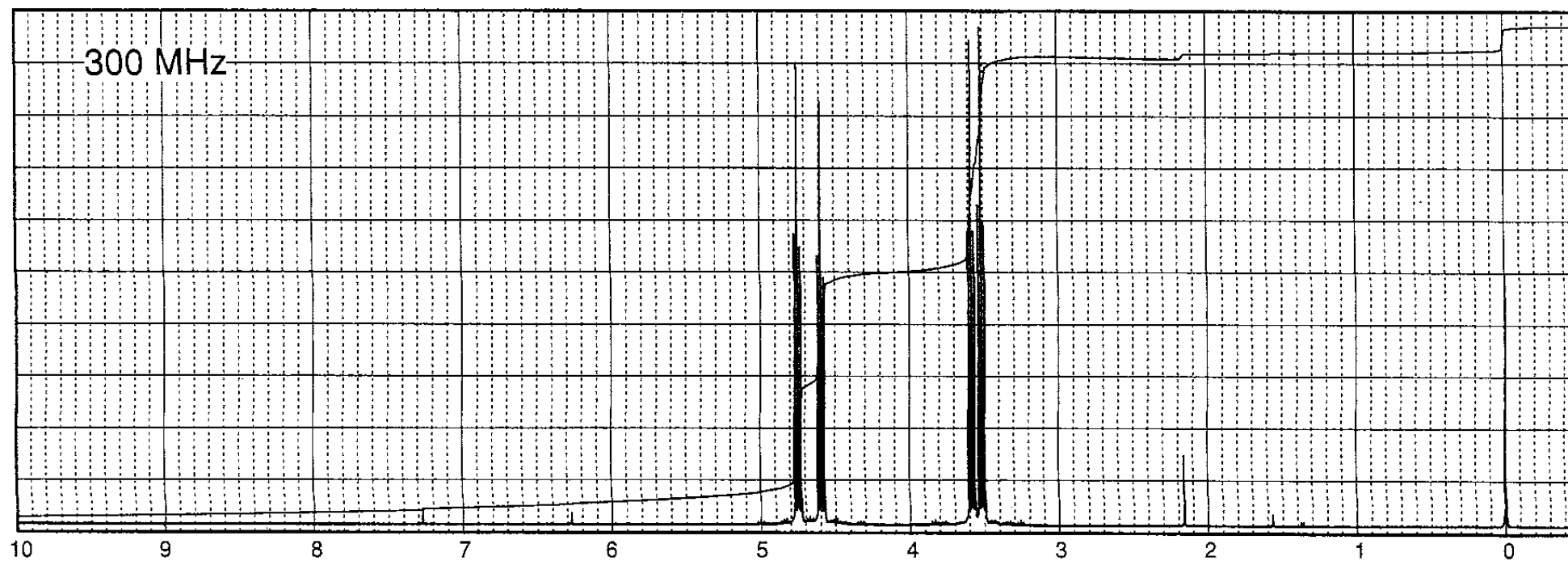
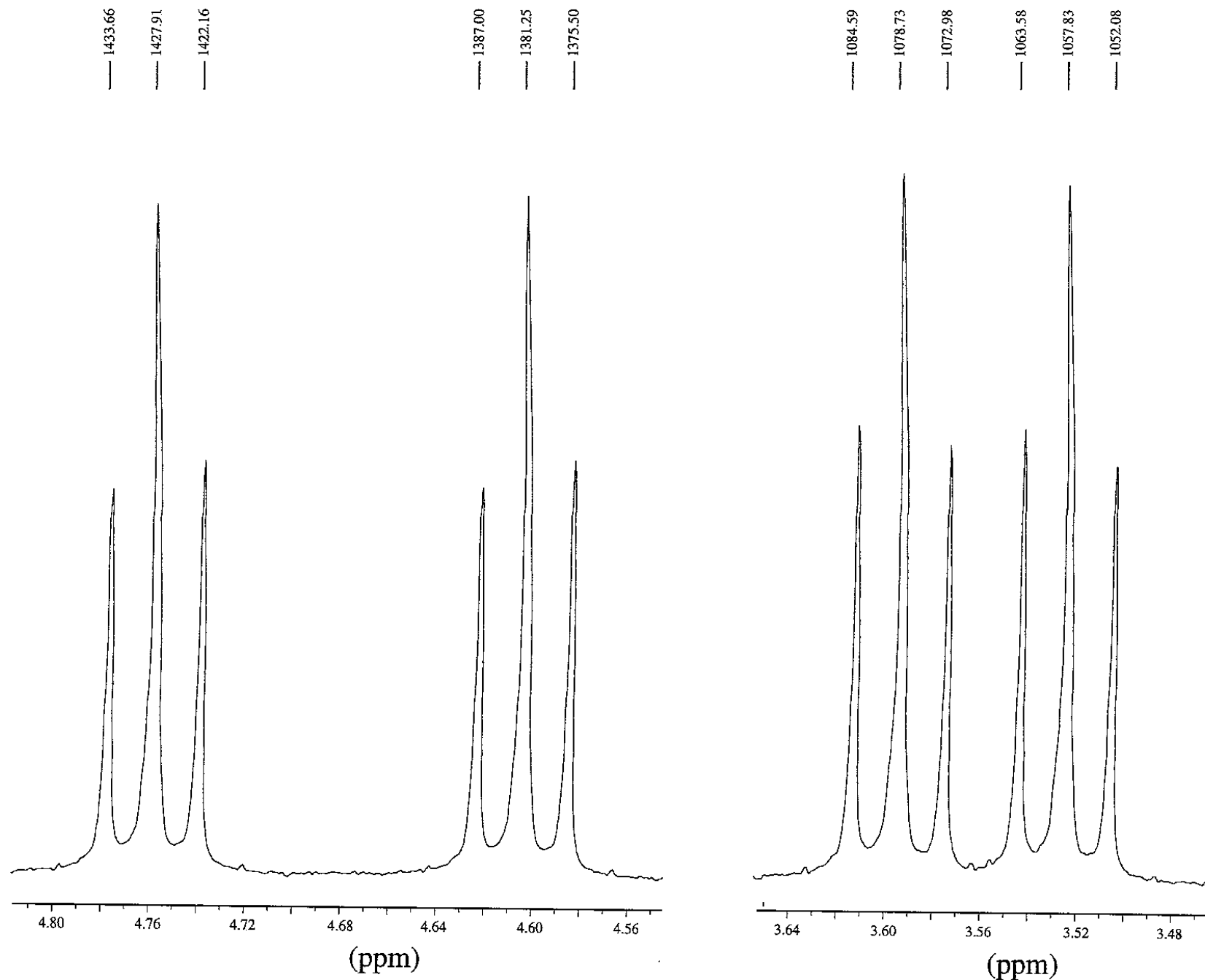


FIGURE 6.23 The ^{13}C proton-decoupled spectrum of tetramethylphosphonium chloride, $(\text{CH}_3)_4\text{P}^+\text{Cl}^-$ (75 MHz).

Q. A compound with the formula $\text{C}_2\text{H}_4\text{BrF}$ has the following NMR spectrum. Draw the structure for this compound. Using the Hertz values on the expansions, calculate the coupling constants. An expansion of the peaks is also provided.



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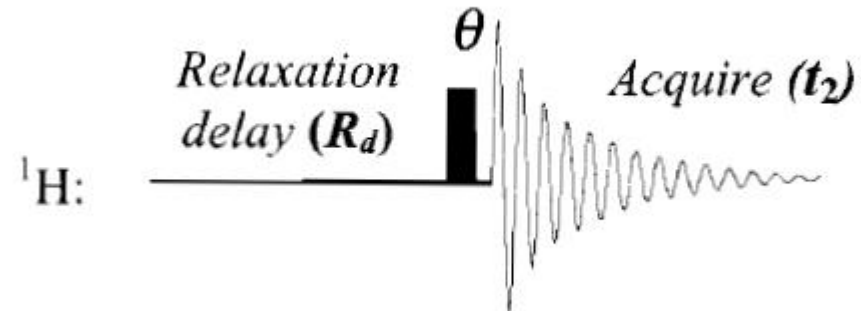
2D NMR Spectroscopy

The Second Dimension

- The development of 2D NMR is a major milestone in the history of NMR spectroscopy
- Has allowed the elucidation of structures of complex macromolecules
- Expanded the application of NMR in fields such as structural biology, proteomics and metabolics

➤ **1D NMR**: only one time-domain; FID is Fourier Transformed into frequency domain

(a)



(b)

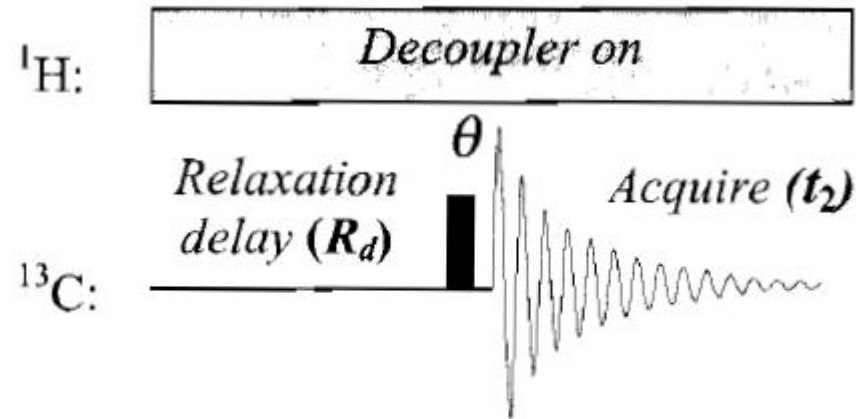


FIGURE 5.2 (a) Pulse sequence for standard 1-D ^1H spectrum. (b) Pulse sequence for a standard (decoupled) ^{13}C spectrum. θ is normally a $(\pi/2)_x$ or a 90° pulse along the x axis. R_d is an equilibration period in the magnetic field before the pulse.

Pulse sequence for simple, decouple and manipulating Nuclear Overhauser Effect (NOE)

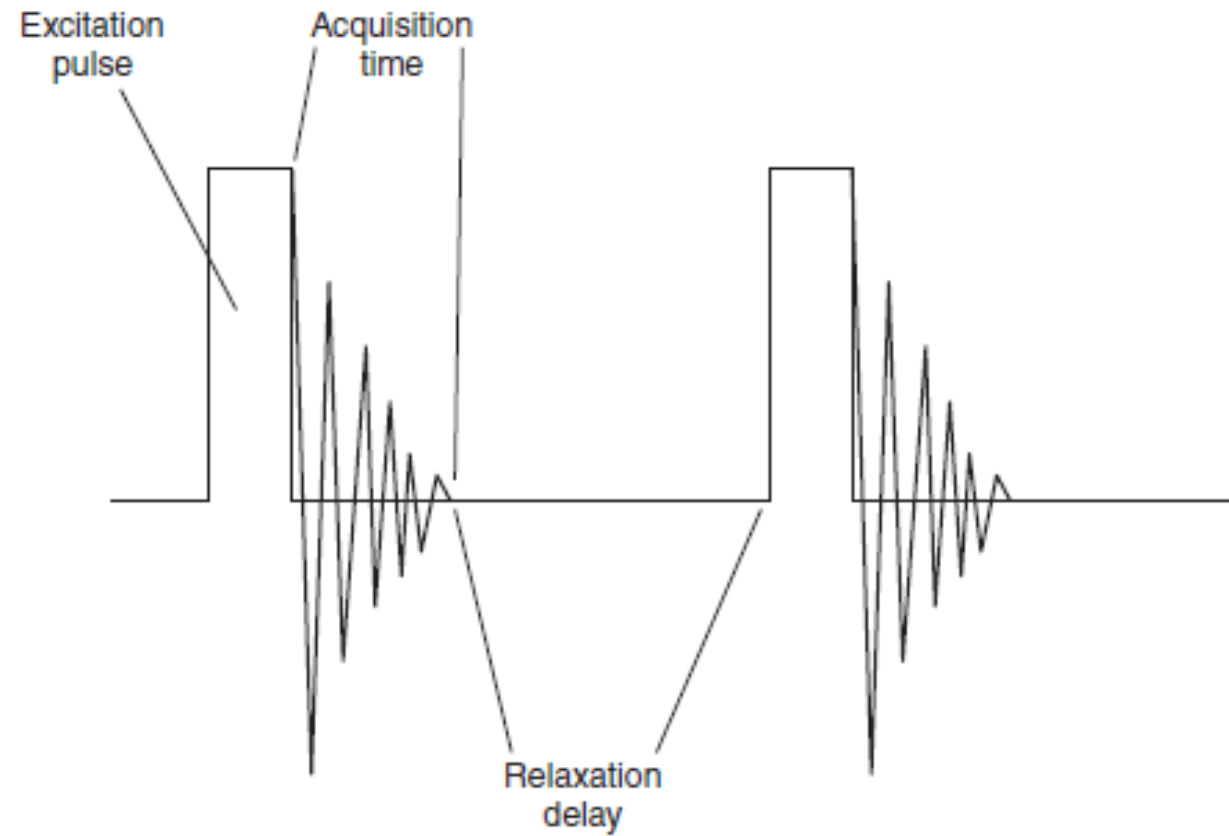


FIGURE 9.1 A simple pulse sequence.

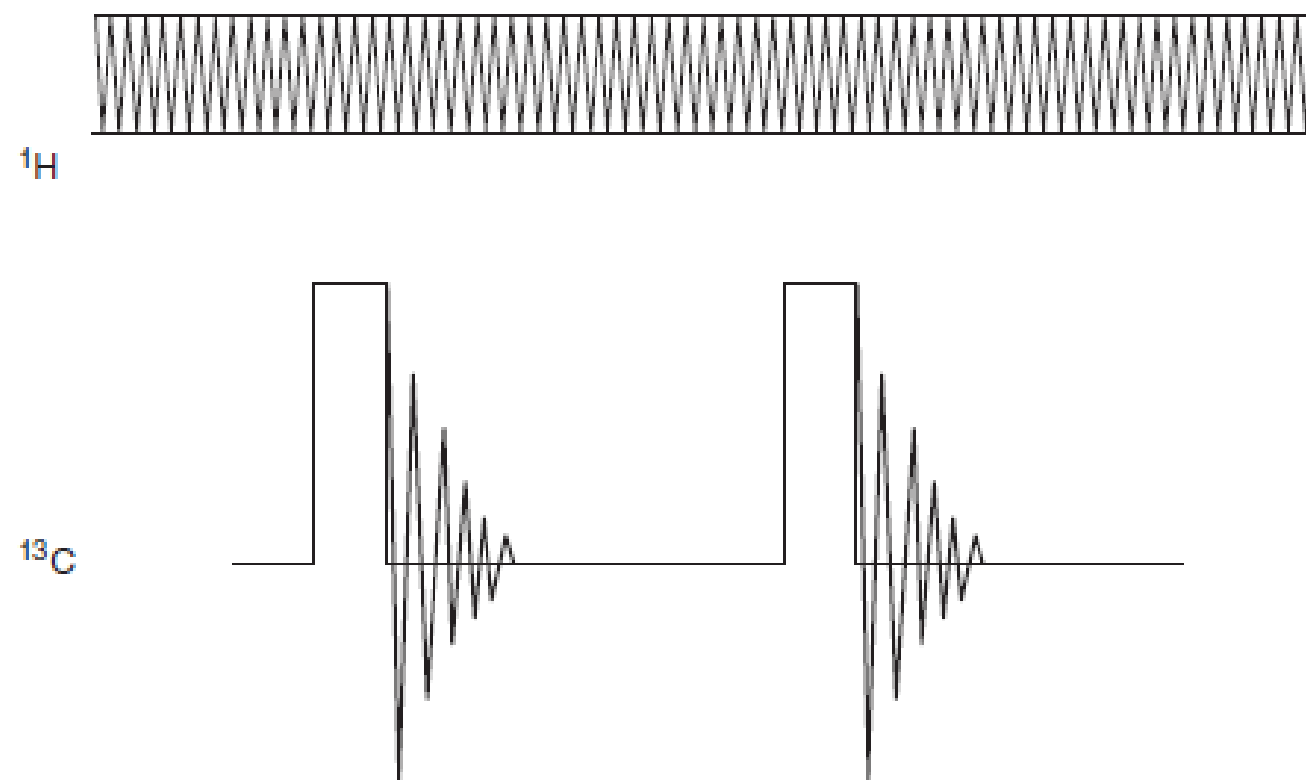
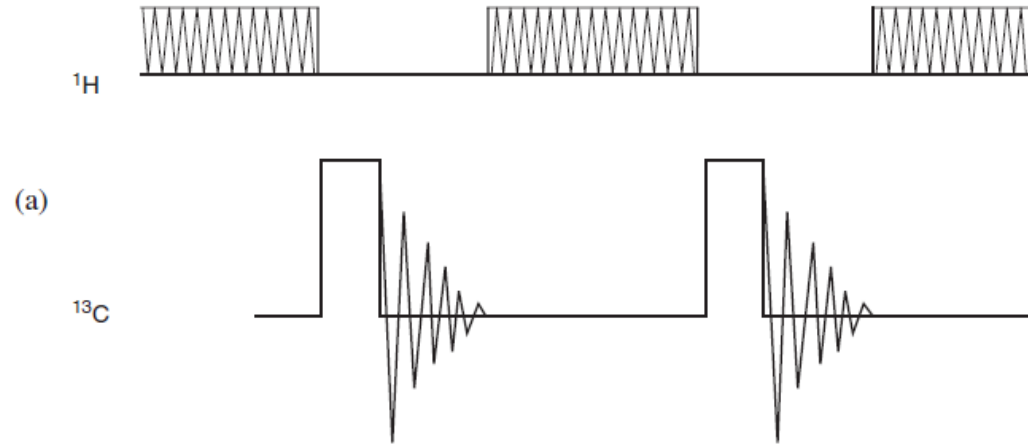


FIGURE 9.2 A proton-decoupled ^{13}C NMR pulse sequence.

NOE-enhanced proton-coupled spectrum or a gated decoupling spectrum



a) The effect of this pulse sequence is to allow the nuclear Overhauser effect to develop while the decoupler is on. Because the decoupler is switched off during the excitation pulse, spin decoupling of the ^{13}C atoms is not observed (a proton-coupled spectrum is seen). The nuclear Overhauser enhancement decays over a relatively long time period; therefore, most of the enhancement is retained as the FID is collected.

NOE-enhanced proton-coupled spectrum or a gated decoupling spectrum

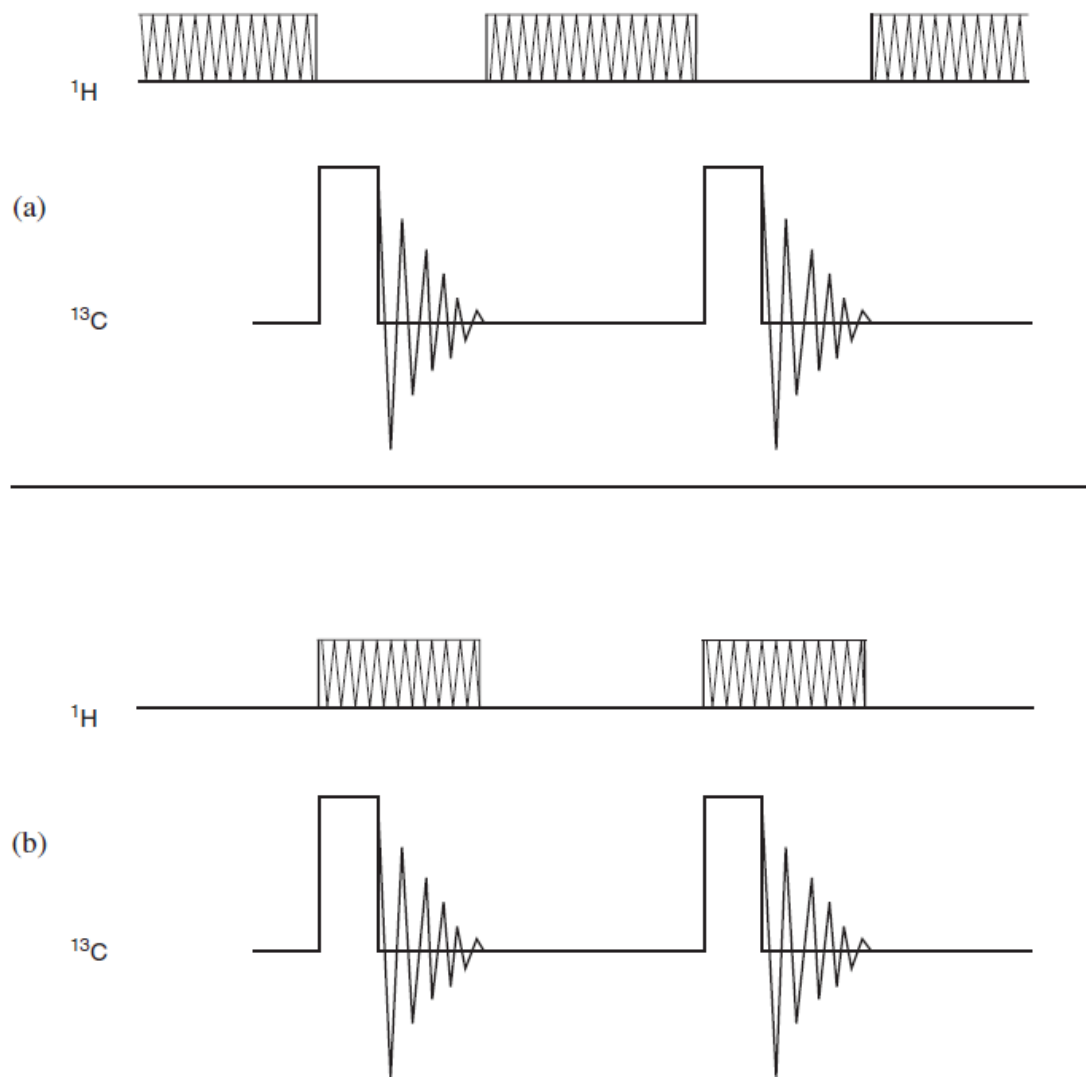
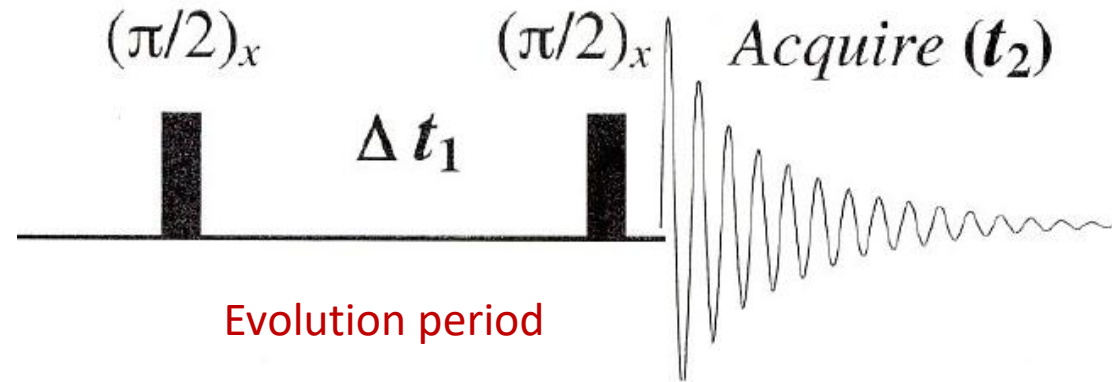


FIGURE 9.3 A simple pulse sequence. (a) A pulse sequence for gated decoupling; (b) a pulse sequence for inverse gated decoupling.

a) The effect of this pulse sequence is to allow the nuclear Overhauser effect to develop while the decoupler is on. Because the decoupler is switched off during the excitation pulse, spin decoupling of the ^{13}C atoms is not observed (a proton-coupled spectrum is seen). The nuclear Overhauser enhancement decays over a relatively long time period; therefore, most of the enhancement is retained as the FID is collected.

b) The opposite result is obtained if the decoupler is not switched on until *the very instant* the excitation pulse is transmitted. Once the FID data have been collected, the decoupler is switched off until the next excitation pulse. This pulse sequence is called **inverse gated decoupling**. The effect of this pulse sequence is to provide a **proton-decoupled spectrum with no NOE enhancement**. Because the decoupler is switched off before the excitation pulse, the nuclear Overhauser enhancement is not allowed to develop. Proton decoupling is provided since the decoupler is switched on during the excitation pulse and the acquisition time.

➤ **2D NMR:** Two time domains



- ❑ In the evolution period the spins are allowed to interact with each other

❑ *Correlation*: connectivity and coupling

- Same group of nuclei
- Nuclei which are coupled

❑ *Nuclear Overhauser effect (NOE)*: spatial proximity

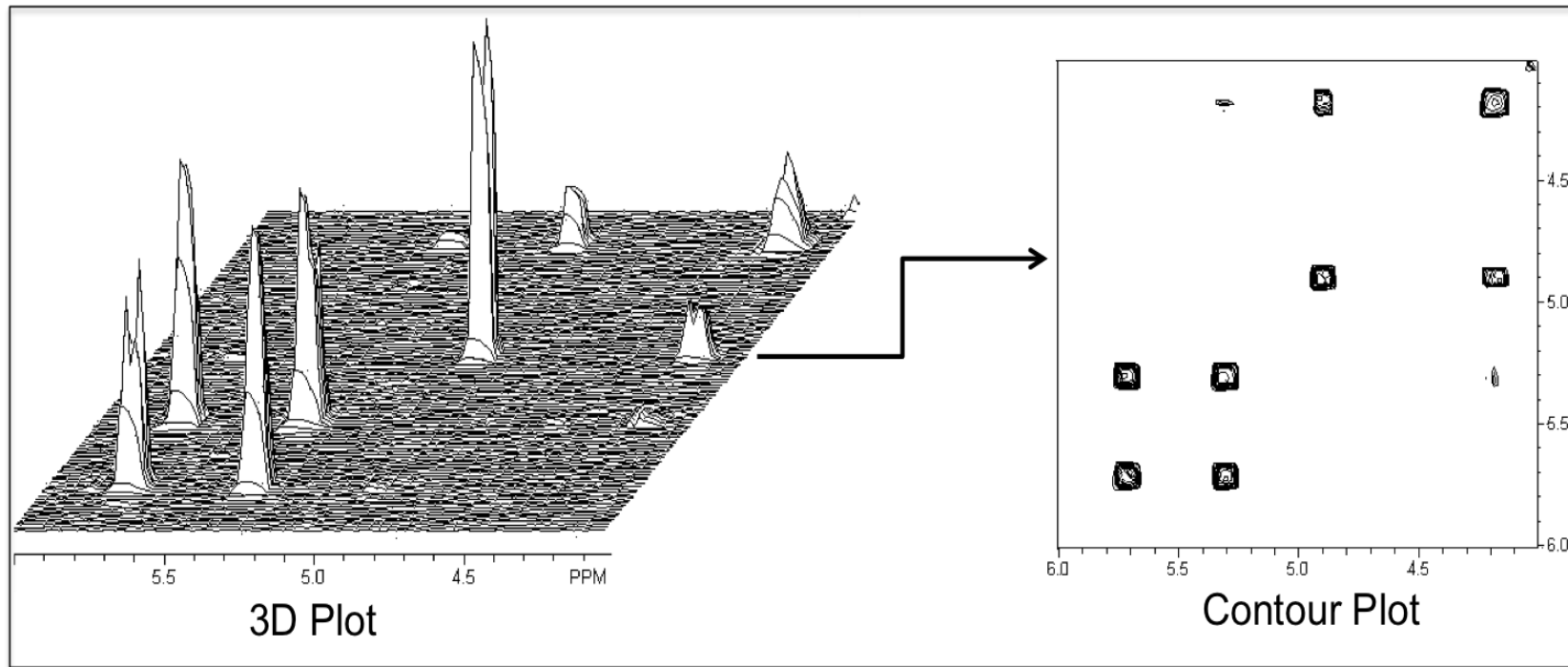
COSY: Correlation spectroscopy (J-coupling)

- ✓ ^1H - ^1H correlation via spin coupling interactions
- ✓ Nuclei must be connected by bonds (not too many bonds!)

HETCOR: Heteronuclear correlation (J-coupling): ^1H - ^{13}C

NOESY: NOE correlation, ^1H - ^1H correlation

- ✓ It correlates nuclei which are close to each other in space
- ✓ But not necessarily coupled to each other
- ✓ Basically works on the principle of NOE



- *Diagonal peaks* have the same frequency coordinate on each axis and appear along the diagonal of the plot. Diagonal peaks correspond to the peaks in a 1D-NMR experiment
- *Cross peaks* have different values for each frequency coordinate and appear off the diagonal; they indicate couplings between pairs of nuclei (as splitting indicates couplings in 1D-NMR)

